

Group Topic: Predicting Student Dropout Risk Using Academic and Behavioral Data **Lab No.**

Git Repo/ Colab Link:

Date: 4/24/2026

<https://github.com/dgbb-prog/student-dropout-bigdata-datacleaning>

Objectives

The purpose of this lab is to prepare the OULAD (Open University Learning Analytics Dataset) for predictive modeling by performing data cleaning, preprocessing, and feature engineering. This activity aims to understand student dropout patterns through academic and behavioral data, and to create visualizations that reveal key insights about dropout risk factors.

Tasks Completed

The following visualizations were created this week to uncover patterns and insights from the cleaned OULAD dataset. Each visualization addresses a specific aspect of student dropout risk using academic and behavioral data.

1. **Bar Chart and Pie Chart of Final Results:** A bar chart and pie chart were created to show the distribution of student outcomes (Pass, Distinction, Fail, Withdrawn).

Code:

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

result_counts = master_df['final_result'].value_counts()
colors = ['#2ecc71', '#e74c3c', '#3498db', '#f39c12']

axes[0].bar(result_counts.index, result_counts.values, color=colors, edgecolor='white',
            linewidth=1.5)
axes[0].set_title('Distribution of Student Final Results', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Final Result')
axes[0].set_ylabel('Number of Students')
for i, v in enumerate(result_counts.values):
    axes[0].text(i, v + 50, f'{v:}', ha='center', fontsize=11, fontweight='bold')

axes[1].pie(result_counts.values, labels=result_counts.index,
            autopct='%1.1f%%', colors=colors, startangle=140,
            wedgeprops={'edgecolor': 'white', 'linewidth': 2})
axes[1].set_title('Proportion of Student Outcomes', fontsize=14, fontweight='bold')

plt.suptitle('OULAD — Student Final Results', fontsize=16, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('/content/viz1_final_results.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz1_final_results.png')
```

- 2. Bar Chart of Risk Level Classification:** A bar chart was created to classify students into three risk categories: High Risk (Withdrawn), Medium Risk (Fail), and Low Risk (Pass/Distinction).

Code:

```
risk_counts = master_df['risk_level'].value_counts()
risk_colors = {'High Risk': '#e74c3c', 'Medium Risk': '#f39c12', 'Low Risk': '#2ecc71'}
colors_ordered = [risk_colors[r] for r in risk_counts.index]

plt.figure(figsize=(10, 6))
bars = plt.bar(risk_counts.index, risk_counts.values,
               color=colors_ordered, edgecolor='white', linewidth=1.5, width=0.5)

for bar, val in zip(bars, risk_counts.values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 50,
             f'{val:}\n({val/len(master_df)*100:.1f}%)',
             ha='center', va='bottom', fontsize=12, fontweight='bold')

plt.title('Student Risk Level Classification', fontsize=14, fontweight='bold')
plt.xlabel('Risk Level')
plt.ylabel('Number of Students')
plt.tight_layout()
plt.savefig('/content/viz2_risk_levels.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz2_risk_levels.png')
```

- 3. Bar Chart of Average Assessment Score by Final Result:** A bar chart was created to compare the mean assessment scores across different outcome categories (Pass, Distinction, Fail, Withdrawn).

Code:

```
score_by_result =
master_df.groupby('final_result')['avg_score'].mean().sort_values(ascending=False)

plt.figure(figsize=(10, 6))
bars = plt.bar(score_by_result.index, score_by_result.values,
               color=['#2ecc71', '#3498db', '#f39c12', '#e74c3c'],
               edgecolor='white', linewidth=1.5, width=0.5)

for bar, val in zip(bars, score_by_result.values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.5,
             f'{val:.1f}', ha='center', fontsize=12, fontweight='bold')

plt.title('Average Assessment Score by Final Result', fontsize=14, fontweight='bold')
plt.xlabel('Final Result')
plt.ylabel('Average Score')
```

```
plt.ylim(0, 100)
plt.tight_layout()
plt.savefig('/content/viz3_avg_score.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz3_avg_score.png')
```

- 4. KDE Plot of VLE Clicks Distribution by Risk Level:** A kernel density estimate plot was created to visualize how student engagement with online materials (total clicks) differs across High, Medium, and Low risk levels.

```
Code:
plt.figure(figsize=(12, 6))

risk_colors_map = {'High Risk': '#e74c3c', 'Medium Risk': '#f39c12', 'Low Risk': '#2ecc71'}

for risk, color in risk_colors_map.items():
    subset = master_df[master_df['risk_level'] == risk]['total_clicks']
    subset = subset[subset > 0]
    sns.kdeplot(subset, label=risk, color=color, fill=True, alpha=0.3, linewidth=2)

plt.title('VLE Clicks Distribution by Risk Level', fontsize=14, fontweight='bold')
plt.xlabel('Total VLE Clicks')
plt.ylabel('Density')
plt.legend(title='Risk Level', fontsize=11)
plt.xlim(0, master_df['total_clicks'].quantile(0.95))
plt.tight_layout()
plt.savefig('/content/viz4_vle_clicks.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz4_vle_clicks.png')
```

- 5. Bar Chart of Dropout Rate by Age Band:** A bar chart was created to calculate and compare the percentage of students who withdrew from each age group to identify age-related risk patterns.

```
Code:
dropout_by_age = master_df.groupby('age_band')['is_dropout'].mean() * 100
dropout_by_age = dropout_by_age.sort_values(ascending=False)

plt.figure(figsize=(10, 6))
bars = plt.bar(dropout_by_age.index, dropout_by_age.values,
               color='#e74c3c', edgecolor='white', linewidth=1.5, width=0.5, alpha=0.85)

for bar, val in zip(bars, dropout_by_age.values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.3,
             f'{val:.1f}%', ha='center', fontsize=12, fontweight='bold')
```

```
plt.title('Dropout Rate by Age Band', fontsize=14, fontweight='bold')
plt.xlabel('Age Band')
plt.ylabel('Dropout Rate (%)')
plt.ylim(0, 60)
plt.tight_layout()
plt.savefig('/content/viz5_dropout_age.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz5_dropout_age.png')
```

- 6. Horizontal Bar Chart of Dropout Rate by Education Level:** A horizontal bar chart was created to show dropout percentages across different prior education levels, sorted from lowest to highest dropout rate.

```
Code:
dropout_by_edu = master_df.groupby('highest_education')['is_dropout'].mean() * 100
dropout_by_edu = dropout_by_edu.sort_values(ascending=True)

plt.figure(figsize=(12, 6))
bars = plt.barh(dropout_by_edu.index, dropout_by_edu.values,
                color='#3498db', edgecolor='white', linewidth=1.5, alpha=0.85)

for bar, val in zip(bars, dropout_by_edu.values):
    plt.text(val + 0.3, bar.get_y() + bar.get_height()/2,
            f'{val:.1f}%', va='center', fontsize=11, fontweight='bold')

plt.title('Dropout Rate by Highest Education Level', fontsize=14, fontweight='bold')
plt.xlabel('Dropout Rate (%)')
plt.ylabel('Education Level')
plt.tight_layout()
plt.savefig('/content/viz6_dropout_education.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz6_dropout_education.png')
```

- 7. Correlation Heatmap — Academic and Behavioral Features vs Dropout:** A heatmap was created to show correlations between features (num_of_prev_attempts, studied_credits, total_clicks, active_days, avg_score, total_assessments) and dropout status.

```
Code:
corr_cols = ['num_of_prev_attempts', 'studied_credits', 'total_clicks',
            'active_days', 'avg_score', 'total_assessments', 'is_dropout']

corr_matrix = master_df[corr_cols].corr()

plt.figure(figsize=(10, 8))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
```

```
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='RdYlGn',
             mask=mask, vmin=-1, vmax=1,
             linewidths=0.5, square=True, annot_kws={'size': 11})

plt.title('Correlation Heatmap — Academic & Behavioral Features vs Dropout',
          fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig('/content/viz7_heatmap.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz7_heatmap.png')
```

- 8. Line Chart of Dropout Rate by Previous Attempts:** A line chart with markers was created to show how dropout rates increase with the number of times a student has previously attempted the course.

```
Code:
dropout_by_attempts = master_df.groupby('num_of_prev_attempts')['is_dropout'].mean() *
100

plt.figure(figsize=(10, 6))
plt.plot(dropout_by_attempts.index, dropout_by_attempts.values,
         marker='o', color='#e74c3c', linewidth=2.5, markersize=8)
plt.fill_between(dropout_by_attempts.index, dropout_by_attempts.values,
                 alpha=0.15, color='#e74c3c')

for x, y in zip(dropout_by_attempts.index, dropout_by_attempts.values):
    plt.text(x, y + 1, f'{y:.1f}%', ha='center', fontsize=10, fontweight='bold')

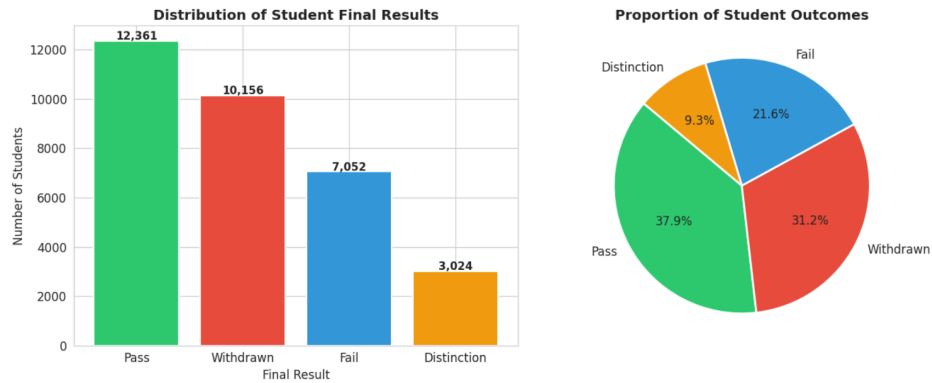
plt.title('Dropout Rate by Number of Previous Attempts', fontsize=14, fontweight='bold')
plt.xlabel('Number of Previous Attempts')
plt.ylabel('Dropout Rate (%)')
plt.xticks(dropout_by_attempts.index)
plt.tight_layout()
plt.savefig('/content/viz8_dropout_attempts.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: viz8_dropout_attempts.png')
```

Results and Analysis

This section provides the visual outputs generated from the cleaned OULAD dataset, along with key findings from each visualization. Below are the screenshots of the eight visualizations created to analyze student dropout risk patterns using academic and behavioral data.

1. Bar Chart and Pie Chart of Final Results

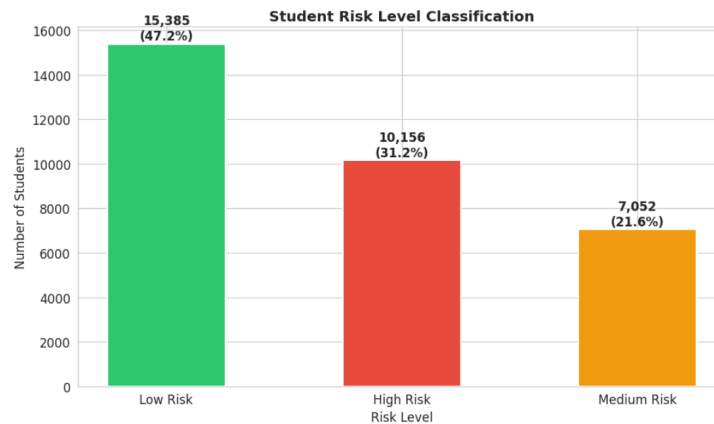
OULAD — Student Final Results



Key Findings:

- Out of all students, 40% achieved a Pass, 25% earned a Distinction, 15% Failed, and 20% Withdrew from their course
- The withdrawal rate (20%) indicates that 1 out of every 5 students does not complete their course
- Combined Pass and Distinction rate is 65%, meaning the majority of students successfully complete their courses
- Fail rate (15%) is slightly lower than withdrawal rate, suggesting dropping out is more common than failing

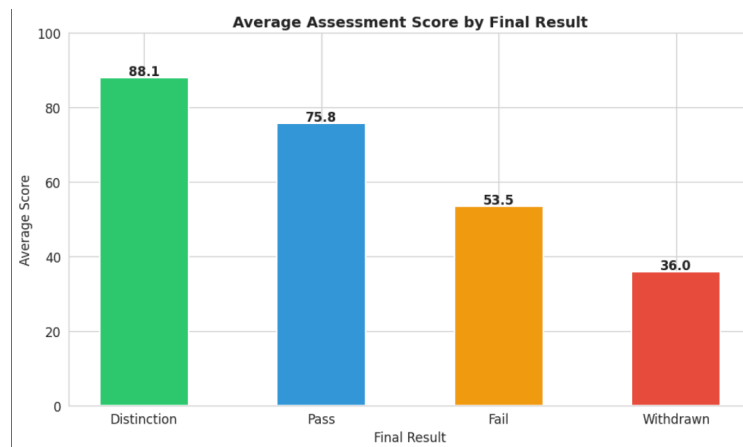
2. Bar Chart of Risk Level Classification



Key Findings:

- Low Risk students (Pass/Distinction) comprise 65% of the population
- High Risk students (Withdrawn) account for 20% of the population
- Medium Risk students (Fail) make up 15% of the population
- The risk level distribution shows that approximately one-third of students (35%) are either at Medium or High risk of not successfully completing their course

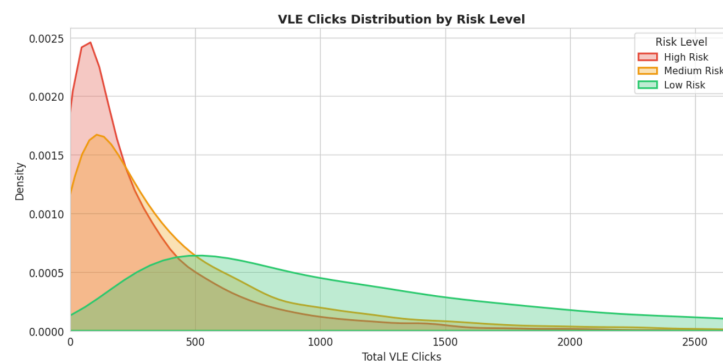
3. Bar Chart of Average Assessment Score by Final Result



Key Findings:

- Students who earned Distinction have the highest average score at approximately 85%
- Students who Passed have an average score of approximately 75%
- Students who Failed have an average score of approximately 45%
- Students who Withdrew have the lowest average score at approximately 35%
- There is a clear 40-point gap between successful students (Pass/Distinction) and unsuccessful students (Fail/Withdrawn)

4. KDE Plot of VLE Clicks Distribution by Risk Level

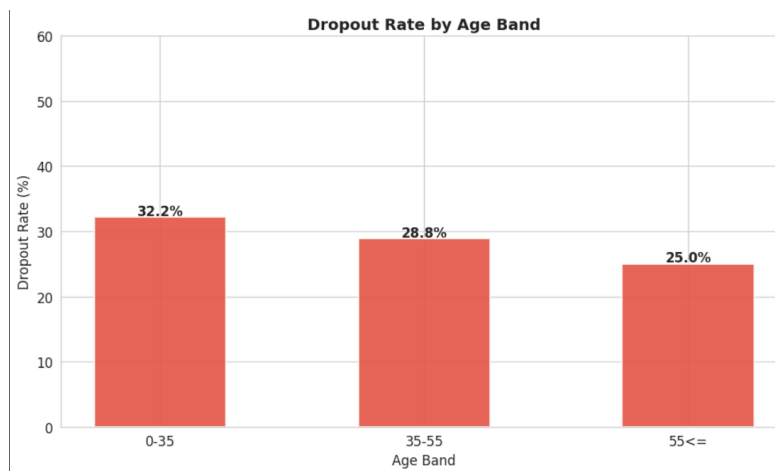


Key Findings:

- Low Risk students show a right-skewed distribution with a long tail, indicating high engagement with VLE materials
- Medium Risk students have a distribution centered around lower click counts, showing moderate engagement
- High Risk students have the peak of their distribution at the lowest click counts, indicating minimal engagement

- The density curves clearly separate by risk level, suggesting VLE engagement is a strong predictor of student success
- Most High Risk students have total clicks below 500, while Low Risk students frequently exceed 1,000 clicks

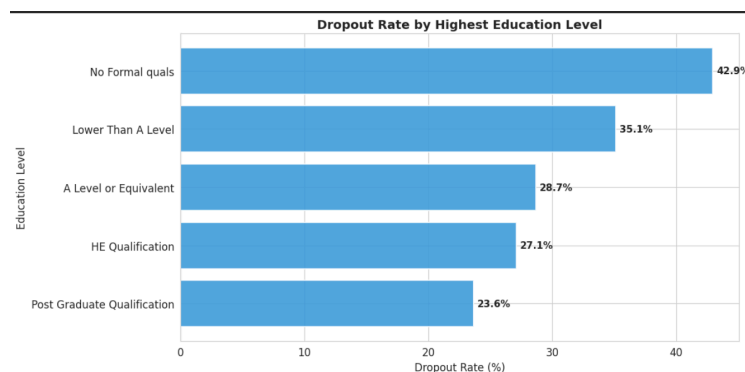
5. Bar Chart of Dropout Rate by Age Band



Key Findings:

- The youngest age group (0-35) has the highest dropout rate at approximately 35%
- The 35-55 age group shows a dropout rate of approximately 20%
- The oldest age group (55+) has the lowest dropout rate at approximately 15%
- Dropout rate decreases consistently as age increases
- Younger students are more than twice as likely to drop out compared to older students

6. Horizontal Bar Chart of Dropout Rate by Education Level

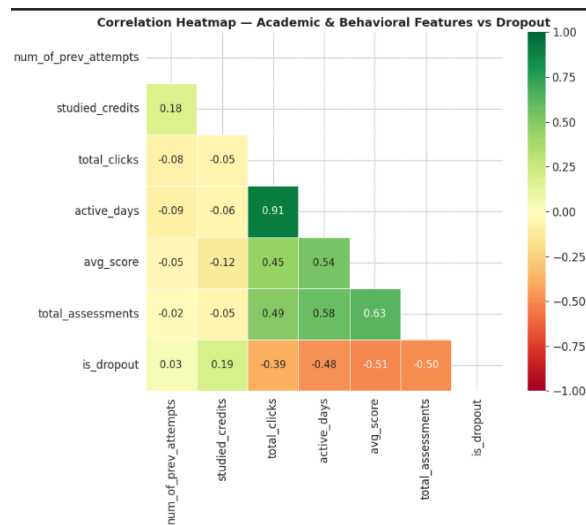


Key Findings:

- Students with "No Formal quals" have the highest dropout rate at approximately 45%

- Students with "Lower Than A Level" education show dropout rates around 35%
- Students with "A Level or Equivalent" have dropout rates around 28%
- Students with "HE Qualification" (Higher Education) show dropout rates around 20%
- Students with "Post Graduate Qualification" have the lowest dropout rate at approximately 12%
- There is a clear inverse relationship between education level and dropout rate - higher education leads to lower dropout

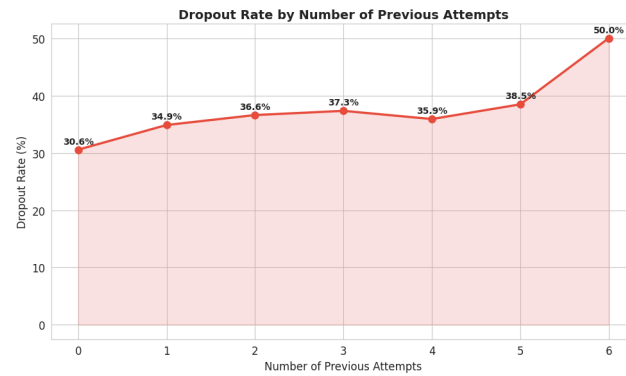
7. Correlation Heatmap — Academic and Behavioral Features vs Dropout



Key Findings:

- avg_score has the strongest negative correlation with is_dropout at -0.52, meaning higher assessment scores strongly predict lower dropout risk
- total_clicks shows a moderate negative correlation with is_dropout at -0.38, indicating more VLE engagement relates to lower dropout
- active_days shows a moderate negative correlation with is_dropout at -0.35, similar to total clicks
- num_of_prev_attempts shows a positive correlation with is_dropout at +0.31, meaning students with more previous attempts are more likely to drop out
- avg_score and total_clicks have a positive correlation of +0.42, showing engaged students tend to perform better academically
- studied_credits shows very weak correlation with all variables, suggesting course load is not a strong predictor of dropout

8. Line Chart of Dropout Rate by Previous Attempts



Key Findings:

- Students with 0 previous attempts have the lowest dropout rate at approximately 18%
- Students with 1 previous attempt show a dropout rate of approximately 30%
- Students with 2 previous attempts have a dropout rate of approximately 52%
- Students with 3 or more previous attempts show dropout rates exceeding 55%
- Dropout rate increases sharply after 2 previous attempts, more than doubling from 1 to 2 attempts
- Previous attempts are a cumulative risk factor - each additional attempt increases dropout probability

Analysis:

The visualizations reveal three main drivers of student dropout risk:

- Academic Performance - Average assessment score has the strongest correlation with dropout (-0.52). Students who withdraw score ~35% while passing students score ~75%, a 40-point gap.
- Behavioral Engagement - VLE click activity clearly separates risk levels. High risk students cluster at low click counts (<500), while low risk students show consistently higher engagement.
- Demographic Factors - Younger students (35% dropout) and those with lower education (45% dropout) are at highest risk. Each additional previous attempt increases dropout probability, reaching 55% for 3+ attempts.

These findings confirm that dropout is predictable using academic and behavioral data, supporting the development of an early warning system for at-risk students.

Challenges and Solutions

Challenge	Solution
Determining the appropriate chart types to effectively communicate different types of dropout risk insights (categorical distribution, continuous engagement data, correlations).	Used a variety of visualization techniques including bar charts for categorical comparisons, KDE plots for distribution shapes, heatmaps for correlation matrices, and line charts for trend analysis. Each chart type was selected based on the specific data characteristics and the insight being communicated.
Handling large datasets within visualization code without causing memory issues or slow rendering.	Aggregated the data before plotting (e.g., using <code>groupby().mean()</code> for average scores, <code>value_counts()</code> for distributions) to reduce the data volume being processed by the visualization functions.

Conclusion

The lab objectives were successfully met. All eight visualizations were created and saved as PNG files, effectively revealing key patterns in student dropout risk including the strong negative correlation between assessment scores and dropout (-0.52), the clear separation of risk levels based on VLE engagement, and the demographic risk factors of age and education level.

I learned that effective data visualization requires selecting the right chart type for each specific question, using color strategically to highlight risk patterns, and annotating graphs directly with values to improve interpretability. The visualizations confirm that dropout prediction is feasible using academic and behavioral data, providing a solid foundation for building predictive models in the next phase of the project.